

The word frequency effect: A review of recent developments and implications for the choice of frequency estimates in German

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Abstract

We review recent evidence indicating that researchers in experimental psychology may have used suboptimal estimates of word frequency. Word frequency measures should be based on a corpus of at least 20 million words that contains language participants in psychology experiments are likely to have been exposed to. In addition, the quality of word frequency measures should be ascertained by correlating them with behavioral word processing data. When we apply these criteria to the word frequency measures available for the German language, we find that the commonly used Celex frequencies are the least powerful to predict lexical decision times. Better results are obtained with the Leipzig frequencies, the dlexDB frequencies, and the Google Books 2000-2009 frequencies. However, as in other languages the best performance is observed with subtitle-based word frequencies. The SUBTLEX-DE word frequencies collected for the present ms are made available in easy-to-use files and are free for educational purposes.

Word frequency is one of the most important variables in experimental psychology. For a start, it is the best predictor of lexical decision times, the time needed to decide whether a string of letters forms an existing word or a made-up nonword in a lexical decision task (Baayen, Feldman, & Schreuder, 2006; Balota, Cortese, Sergent-Marshall, Spieler, & Yap, 2004; Keuleers, Diependaele, & Brysbaert, 2010b; Yap & Balota, 2009). As Murray and Forster (2004, p. 721) concluded: “Of all the possible stimulus variables that might control the time required to recognize a word pattern, it appears that by far the most potent is the frequency of occurrence of the pattern ... Most of the other factors that influence performance in visual word processing tasks, such as concreteness, length, regularity and consistency, homophony, number of meanings, neighborhood density, and so on, appear to do so only for a restricted range of frequencies or for some tasks and not others”.

The importance of word frequency

To illustrate the importance of word frequency, we downloaded the lexical decision times and the word features from the Elexicon Project (Balota et al., 2007; available at <http://elexicon.wustl.edu/>). This database contains lexical decision times and naming times for 40,481 English words, together with information about over 20 word variables, including:

- Frequency (log subtitle based word frequency or SUBTLEX; see below for more information)
- Orthographic length of the word (number of letters)
- Number of orthographic, phonological, and phonographic neighbors (i.e., the number of words that differ in one letter or phoneme from the target word, either with or without the exclusion of homophones)

- Orthographic and phonological distance to the 20 closest words (i.e., the minimum number of letter substitutions, deletions, or additions that are needed to turn a target word into 20 other words); both unweighted or weighted for word frequency
- The mean and sum of the bigram frequencies (i.e., the number of words containing the letter pairs within the target word); either based on the total number of words or limited to the syntactic class of the target word
- The number of phonemes and syllables of the word
- The number of morphemes in the word

When all these variables are entered in a stepwise multiple regression analysis, the most important variable to predict lexical decision time is word frequency, accounting for 40.5% of the variance (Table 1). The second most important variable is the orthographic closeness of the target word to the 20 nearest English words, called the Orthographic Levenshtein Distance (Yarkoni, Balota & Yap, 2008). It accounts for 12.9% additional variance. The unique contribution of the third variable, the number of syllables, already drops to 1.2%, and the summed contribution of the remaining variables amounts to a mere 2.0%. Other authors also reported that the unique contribution of most variables studied in psycholinguistics (such as imageability, age of acquisition, familiarity, spelling-sound consistency, family size, number of meanings, ...) is less than 5% of variance, once the effect of word frequency is partialled out (Baayen et al., 2006; Balota et al., 2004; Cortese & Khanna, 2007; Yap & Balota, 2009).

Insert Table 1 about here

Word frequency is important in word naming as well, but less so. For monosyllabic words the nature of the first phoneme is more important (Balota et al., 2004; Cortese & Khanna, 2007; Yap & Balota, 2009). Finally, word frequency is a variable of importance in memory research as well. In this research, participants first study a list of words and are later required to recall the stimuli or to discriminate them from lures (new items). Interestingly, here the pattern of results depends on the task: Low-frequency words in general are more difficult to recall but lead to better performance in a recognition task (i.e., they result in higher d' values as calculated by signal detection theory; e.g., Cortese, Khanna, Hacker, 2010; Gregg, Gardiner, Karayianni, & Konstantinou, 2006; Higham, Bruno, & Perfect, 2010; Yonelinas, 2002; see also Kang, Balota, & Yap, 2009, for an example of how this reverse frequency effect can be attenuated by context). Because of the importance of word frequency, no study in word recognition or memory research can afford to leave out this variable.

Quality control of word frequency measures

Given the weight of word frequency for research in experimental psychology, one would expect a thriving literature on the quality of the frequency measures used. Surprisingly, this is not the case. Researchers seem to use whatever frequency measure they can get their hands on, with a preference for some “classic” lists. For instance, most research in English has been based on the Kucera and Francis (KF; 1967) word frequency measure (Brysbaert & New, 2009). Four other measures occasionally used are Celex (Baayen, Piepenbrock, & van

Rijn, 1995), Zeno et al. (Zeno, Ivens, Millard, & Duvvuri, 1995), HAL (Balota et al., 2007; Burgess & Livesay, 1998), and the British National Corpus (Leech, Rayson, & Wilson, 2001).

The continued use of KF is surprising, given that the few studies assessing its quality relative to the other measures have been negative. Burgess and Livesay (1998) correlated KF and HAL frequencies with lexical decision times for 240 words, and reported correlations of $-.52$ ($R^2 = .27$) and $-.58$ ($R^2 = .34$) respectively. The HAL word frequency measure was based on a corpus of 131 million of words downloaded from internet discussion groups. Similar conclusions were reached by Zevin and Seidenberg (2002), Balota et al. (2004), and Brysbaert and New (2009): Of all frequency measures tested KF was the worst, followed by Celex. The difference in variance explained could easily amount to 10%, which is substantial given the limited percentages of variance accounted for by most word features. This leaves open the possibility that a number of effects reported in the literature may be invalid, due to improper control of word frequency. Brysbaert and Cortese (2011), for instance, examined the impact of a better word frequency measure on the influences of familiarity (assessed by asking participants how familiar they were with the words) and age of acquisition (assessed by asking participants at what age they learned the various words). Brysbaert and Cortese investigated the impact on the lexical decision times for 2,336 monosyllabic English words. When all three variables (frequency, familiarity, age of acquisition) were entered into the regression analysis, the percentage of variance explained was 52%, independent of the word frequency measure. With the KF measure, however, only 32% was explained by word frequency, 18% by age of acquisition, and 2% by familiarity. In contrast, with the SUBTLEX frequency (subtitle based word frequency), the shares were respectively 44%, 7%, and 1%.

Even though the effect of age of acquisition remained significant, its importance was seriously reduced with a better frequency measure; that of familiarity became negligible.

As part of their validation studies, Brysbaert and New (2009; see also New, Brysbaert, Veronis, & Pallier, 2007) also made a surprising discovery. They found that word frequencies based on subtitles from films and television series consistently outperformed word frequencies based on written documents. They called the new word frequency measure SUBTLEX frequencies. The better performance of SUBTLEX frequencies has since been replicated in Chinese (Cai & Brysbaert, 2010), Dutch (Keuleers, Brysbaert, & New, 2010a), Greek (Dimitropoulou, Dunabeitia, Aviles, Corral, & Carreiras, 2010), and Spanish (Cuetos, Glez-Nosti, Barbon, & Brysbaert, 2011). This raises the question what aspects of word frequency measures are important for their quality.

Variables influencing the quality of word frequency estimates

Given the consistent quality differences between word frequency estimates, it becomes interesting to know which variables must be taken into account. The following have been identified as important.

Size. A first variable determining the quality of a word frequency estimate is, not surprisingly, the size of the corpus: A corpus of 10 million words is better than a corpus of one million words. However, the logic behind this factor is not quite as most researchers understand it. Large corpora are better than small corpora, not because all estimates become better but because the estimates of the very low frequency words are more reliable. Recent analyses with the Elexicon data (Balota et al., 2007) and other large

databases of lexical decision times, such as the French Lexicon Project (Ferrand, New, Brysbaert, Keuleers, Bonin, Meot, Augustinova, & Pallier, 2010) and the Dutch Lexicon Project (Keuleers et al., 2010b), have shown that nearly half of the word frequency effect is situated below frequencies of 2 per million (pm; see Figure 1).

Insert Figure 1 about here

As can be seen in Figure 1, the word frequency curve in lexical decision is nearly flat between frequencies of 50 pm ($\log_{10} = 1.7$) and 40,000 pm ($\log_{10} = 4.6$). In contrast, almost the complete frequency effect is due to frequencies below 10 pm ($\log_{10} = 1$) with a huge effect, for instance, between frequencies of .1 pm ($\log_{10} = -1$) and 1 pm ($\log_{10} = 0$). So, one reason why the KF frequencies are explaining less variance is the fact that this measure is based on a corpus of 1 million words only. This makes it impossible to make fine-grained distinctions between the low-frequency words. The widespread use of the KF measure arguably also has prevented authors from discovering the importance of frequency differences at the low end of the distribution. Indeed, a review of studies manipulating word frequency indicates that nearly all researchers defined low frequency rather loosely as values below 5 pm and invested most of their energy in finding the best possible high-frequency words (expecting large processing differences between words with frequencies of 50 pm and 300 pm).

Incidentally, the fact that word frequency matters for words encountered with a frequency of less than .1 pm means that the frequency effect really is a learning effect. Assuming that

20-year olds have been exposed to a maximum of 1.4 billion words in their life (200 per minute * 60 minutes * 16 hours a day * 365.25 days * 20 years), a frequency of .1 pm means that the word has been encountered only 140 times in the entire life of a typical participant. The fact that such a word is processed significantly faster than a word with a frequency of .02 pm (encountered 28 times) implies that each exposure to a word matters for the word frequency effect and that the effect is not limited to words processed thousands of times.

Although the size of the corpus must be taken into account when assessing the quality of a frequency measure, further scrutiny has revealed that its importance rapidly diminishes as the corpus becomes larger. Whereas there is a clear advantage of 10 million over 1 million, there is virtually no gain above sizes of 20-30 million (Brysbaert & New, 2009; Keuleers et al., 2010a). As we will see below, although larger sizes mean better estimates of the very low frequency words, analyses of existing billion word corpora suggest that they usually perform less well than smaller corpora, because they are less representative of the language read and heard by participants in psycholinguistic research. This brings us to the next variable: the language register measured by the corpus.

Language register. If the word frequency effect is a practice effect, as indicated above, the quality of the frequency measure will depend on the extent to which the materials in the corpus mimic the language typical participants in experimental psychology have been exposed to. Most of the time, these are undergraduate students in psychology. Up to recently, there was not much choice of frequency lists, and psychologists had to do with the few lists compiled for them. However, nowadays with the massive availability of language in digital form researchers can become more selective.

The impact of the type of language sampled became clear when some very large scale corpora were analyzed (Brysbaert & New, 2009; Brysbaert, Keuleers, & New, 2011). For instance, Google recently published the Google Ngram Viewer, which includes word frequency estimates based on the gigantic digitized Google Books corpus including millions of books published since 1500 (Michel et al., 2011; available at <http://ngrams.googlelabs.com/>). When we used the word frequencies on the basis of the American English subcorpus (for a total of 131 billion words!) and correlated them with the Elexicon lexical decision times as in Table 1, the correlation was only $r = -.543$ (or $R^2 = .295$), well below the percentage of variance explained by SUBTLEX (Table 1). Performance of the Google Ngram estimates was slightly better when the corpus was limited to fiction books (i.e., the English Fiction subcorpus, 75 billion words). Then the correlation increased to $-.576$ ($R^2 = .332$) despite the smaller size of the corpus.

The findings with Google Ngram Viewer illustrate a typical problem faced by psycholinguistic researchers: Corpora compiled by linguistics or other organizations usually are representative for the type of materials published, but not (necessarily) for the type of language read by participants of psycholinguistic experiments. In general, non-fiction texts tend to be overrepresented in written corpora.

On the basis of analyses with the Elexicon data, Brysbaert and New (2009) recommended three good sources of word frequency estimates. The first, and most important, consists of word frequencies based on subtitles of popular films and television series. These consistently explain most of the variance in word recognition times when based on a corpus of at least 20

million words. Their good performance presumably is due to the fact that people watch quite a lot of television in their life, and to the fact that the language on television is more representative of everyday word use in social situations. A second interesting source consists of books used by children in primary and secondary school (as measured by Zeno et al., 1995, for English). The suitability of these data arguably has to do with the fact that all university students studied these books (or similar ones) and with the fact that early acquired words have a processing advantage over later learned words (e.g., Izura, Pérez, Agallou, Wright, Marin, Stadthagen-Gonzalez, & Ellis, 2010; Monaghan & Ellis, 2010; Stadthagen-Gonzales, Bowers & Damian, 2004). The frequency of word use in childhood tends to be overlooked when a corpus is exclusively based on materials written for an adult audience. An interesting extension in this respect will be to see whether subtitle frequencies based on television for children are an interesting addition to the SUBTLEX frequencies. Finally, the traditional written frequencies also seem to add one or two percent of variance, especially when they are based on popular sources, such as widely read newspapers and magazines or internet discussion groups. Brysbaert and New (2009) showed that a composite frequency measure based on SUBTLEX, Zeno et al., and HAL explained most of the variance in the Elexicon data.

Variation in time. Finally, word use also shows variation in time. New words are introduced and increase in popularity, other words decrease in use. As a result, word frequency measures become outdated after some time. An example of this was reported by Brysbaert and New (2009, Footnote 6). They observed that word frequency estimates based on pre-1990 subtitles compared to post-1990 subtitles explained 3% less of the variance in lexical decision times for young participants (20 years) but 1.5% more for old participants (70+

years). Similarly, the Google Ngram Viewer estimates are better when they are based on books published after 2000 rather than on all books in the corpus. For instance, the correlation between Google Fiction 2000-2009 and the Elexicon lexical decision times was $r = -.607$ ($R^2 = .368$). It should be taken into account, however, that a large part of the 2000+ advantage is due to the fact that a more representative sample of books seems to have been included in the Google Books corpus since 2000 than before (Brysbaert et al., 2011). The limited shelf life of word frequencies is likely to be one of the reasons why the KF and Celex frequencies explain less variance than more recent frequency counts.

What about the German language?

The status of word frequency measures in the German language is very similar to that in English, although German in general seems to trail behind English, Dutch, and French, rather than take the lead, as was the case in the nineteenth century. As a matter of fact, the first known word frequency list based on word counting was published in German by Kaeding (1897/1898; see Bontrager, 1991). It was based on a corpus of 11 million and compiled for stenographers. In addition to word frequencies, Kaeding's list contained frequencies of syllables and letters. Unfortunately, to our knowledge this list has not been used in innovative research. Although Cattell did some research on word reading in the nineteenth century in Wundt's laboratory, the first study really addressing the influence of word frequency on word recognition was Howes and Solomon (1951), run shortly after the publication of Thorndike and Lorge's (1944) list of English word frequencies.

Word frequency research in German began after the Max Planck Institute of Nijmegen published the German Celex word frequencies (Baayen, Piepenbrock, & van Rijn, 1995;

available at <http://celex.mpi.nl/>). This list was based on a corpus of 5.4 million German word tokens from written texts such as newspapers, fiction and non-fiction books, and 600,000 tokens from transcribed speech. The written part was a combination of the Mannheimer Korpus I & II and the Bonner Zeitungskorpus 1, while the spoken part was known as the Freiburger Korpus. The Celex word frequency list has been used by most German researchers in the past 15 years. A further interesting aspect is that it also gives information about the syntactic roles of the words and the lemma frequencies. The latter are the summed frequencies of all inflections of a word (e.g., the singular and plural forms of nouns, and the various inflections of verbs).

An alternative to the Celex frequency list has been compiled since the early 2000s at the University of Leipzig and is known as the Leipzig Wortschatz Lexicon (available at <http://wortschatz.uni-leipzig.de/>). This corpus initially contained 4.2 million spoken and written words. One million words came from transcribed speech, newspapers, literature, and academic texts each. The remaining 200 thousand words came from instructional language. This list formed the basis for a frequency based German dictionary (Jones & Tschirner, 2006) and has constantly been updated with new materials. By 2007, the corpus expanded to 30 million words, mainly based on newspapers (Biemand, Heyer, Quasthoff, & Richter, 2007).¹

A recent addition is the dlexDB database (available at www.dlexdb.de) compiled by the University of Potsdam and the Berlin-Brandenburg Academy of Science (Heister et al., 2011). This list is based on the *Digitales Wörterbuch der deutschen Sprache*, comprising over 100 million word tokens, roughly equally distributed over fiction, newspapers, scientific

¹ The frequencies reported below were based on a corpus of 49 million words made available by the authors to M. Conrad in 2009.

publications, and functional literature. Like the Celex database, the dlexDB database also provides information about lemma frequencies and the syntactic roles (parts of speech) taken by words.

As part of the Google Ngram Viewer project, Google made available German word frequencies based on their occurrences in the digitized Google Books corpus (available at <http://ngrams.googlelabs.com/>). The most impressive aspect of this source is the size of the corpus: 37 billion tokens, of which 30.1 billion are words or numbers (most of the remaining tokens are punctuation marks). A further interesting aspect of the Google database is that the frequencies are separated as a function of the year in which the books were published. This allows researchers to see changes in word use. For the purpose of the present study, we calculated separate frequencies for books published in the 1980s (3.3 billion words), 1990s (2.9 billion words), and 2000s (6.05 billion words). Unfortunately, Google does not divide the German corpus into a fiction vs. non-fiction part, as is the case for English.

Finally, given the ease to calculate words in digital files nowadays, it is rather straightforward to establish SUBTLEX frequencies for German as well. We downloaded subtitles of 4,610 films and television series from www.opensubtitles.org and cleaned them for unrelated materials (e.g., information about the film and the subtitles). This gave a corpus of 25.4 million words. For each word we counted how often it occurred starting with a lowercase letter and with an uppercase letter. A particularity of the German spelling is that nouns begin with a capital. So, words used both as a noun and another part of speech have forms starting with a capital or not, also depending on whether they are the first word of a sentence or not (e.g., the word “achtjährige” can be used both as a noun and as an

adjective). To preserve information about the capitalization, the raw stimulus file retained separate entries for words starting with an uppercase and a lowercase letter. In the cleaned stimulus list (used for the present analyses), entries were summed over letter cases (see below for a more detailed description of the information included in the raw and the cleaned version of SUBTLEX-DE).

Testing the quality of the German word frequency measures

To assess the quality of frequency measures, one needs word processing times, preferentially lexical decision times given that this variable is most sensitive to word frequency (Balota et al., 2004). Indeed, research on the validity of word frequency measures started once experimenters had collected word processing times for large numbers of words. As summarized above, much use has been made of the 40 thousand words included in the Elexicon Project (Balota et al., 2007). However, research by Burgess and Livesay (1998), New et al. (2007), and Cai and Brysbaert (2010) suggests that such a large size is not needed. Good results can already be obtained with a few hundred RTs sampled from the entire frequency range.

Below, we test the quality of the different word frequency measures by correlating them with RTs collected in three series of lexical decision experiments. We not only examine the word form frequencies, but also the lemma frequencies of Celex and dlexDB. Lemma frequency is the sum of all inflected forms of a word. For instance, the word “abgelehnt” is an inflected form of the verb “ablehnen” and in addition is sometimes used as an adjective. The lemma frequency of the word “abgelehnt”, therefore, includes all inflected forms of the verb and the adjective. An enduring question in psycholinguistics is to what extent low-

frequency inflected forms are recognized as a unit or are processed by means of a parsing process decomposing the complex word into a stem and suffix. If the latter is true, lemma frequencies may be a better estimate of word exposure. For English, Brysbaert and New (2009) found that word form frequencies were as good as lemma frequencies. However, the number of inflected forms is substantially smaller in English than in German.

Dataset 1: 455 words presented in three lexical decision tasks

The first dataset we used to validate the various word frequency measures involved 455 words. The words were presented in three lexical decision experiments with different types of non-words.

Participants. Participants were three groups of 29 undergraduate students from the Freie Universität Berlin. All had normal or corrected to normal vision and were naïve with respect to the research hypothesis. Their participation was rewarded with course credits and a financial compensation in one of the experiments where EEG was recorded (see below).

Stimuli. The word stimuli consisted of 455 words, which were selected to examine effects of emotional valence and arousal in visual word recognition (Recio, Conrad, Hansen, Schacht, Baier, & Jacobs, in preparation; see also Vö, Conrad, Kuchinke, Hartfeld, Hofmann, & Jacobs, 2009). Three different types of non-words were presented to test for potential effects of this variable (see Grainger & Jacobs, 1996). In the first experiment, easy non-words were used (i.e., non-words containing low frequency letter combinations that made pronunciation difficult although always possible). In the second experiment, largely the same non-words were used but they were supplemented by 53 pseudohomophones (these are

non-words that sound like words). Finally, in the third experiment, difficult pseudowords were used. These pseudowords were made of high-frequency letter combinations and were easily pronounceable. All nonwords were matched to the words on length, syllable number and initial capitals.

Method. Before the test session 10 practice trials were administered. The test session itself consisted of 910 trials (half words and half nonwords) presented in lowercase (Courier 24) with initial capitals for nouns. The time line of a trial was as follows: Items were presented in the center of the screen after a fixation cross (500ms) until responses were given, followed by an intertrial interval of 1500 ms. Participants responded with their right hand to the word trials and with their left hand to the non-word trials in Experiments 1 and 3. Type of responses to be given was balanced between participants in Experiment 2 where EEG was recorded. The EEG-recording was unrelated to the goals of the current study. Each participant got a different, random permutation of the stimulus list.

Results. The mean reaction time for words of Experiment 1 was $RT1 = 636$ ms (percentage of errors: $PE1 = 4.2\%$); for Experiment 2 the data were $RT2 = 616$ ms and $PE2 = 3.5\%$; for Experiment 3 they were $RT3 = 686$ ms, $PE3 = 5.0\%$. The better performance in Experiment 2 than in Experiment 1 was unexpected, given that Experiment 2 contained pseudohomophone non-words (which usually are difficult to reject). The most likely reason for the better performance in Experiment 2 is that participants devoted more energy to the task, because EEG was recorded.

When correlating word frequency measures with performance variables, a recurrent problem is what to do with words that do not occur in a frequency database. For Celex 48 words of the present experiments were missing, for Leipzig 2 words, for SUBTLEX 9 words, and for dlexDB and Google no words. One solution used in the past is to limit the analyses to the words for which there are data in all cells (e.g., Keuleers et al., 2010b). However, this misses the point that some databases have more empty cells than others. Therefore, we decided to give all missing frequencies a value of 0 and to calculate the log frequencies on the basis $\log(\text{frequency} + 1)$ or $\log(\text{frequency per million} + 1/N)$, in which N = the number of words in the corpus expressed in millions (i.e., = 25.4 for SUBTLEX, 100 for dlexDB, and 30,840 for Google Ngram).² This is the procedure many researchers use in practice when they select stimulus materials (i.e., they assume that words not found in the frequency list have a frequency of 0). The surface frequency of the stimuli was rather low: mean $\log_{10}(\text{Google Ngram per million} + 1/30840) = .32$ (equivalent to a frequency of 2 per million), $SD = .876$. The low frequency on average is good because the frequency effect is strongest in this range (Figure 1).

Table 2 shows the correlations between the dependent variables of the three experiments (RT1-RT3, PE1-PE3) and the Celex word form frequencies (CELEXwf), Celex lemma frequencies (CELEXlem), Leipzig word form frequencies (Leipzig), dlexDB word form frequencies (dlexDBwf), dlexDB lemma frequencies (dlexDBlem), Google Ngram (Google),

² It may be good to know that the addition of 1 or $1/N$ tends to decrease the percentage of variance accounted for in the words having a frequency in the database. This is because the addition of a constant attenuates the differences between the very low frequency words.

Google Ngram 1980-1989 (Google80), Google Ngram 1990-1999 (Google90), Google Ngram 2000-2009 (Google00), and SUBTLEX word form frequencies.

Insert Table 2 about here

Five interesting findings emerge from Table 2:

1. The Celex frequencies are the least good measure. Based on the Hotelling-Williams test (Williams, 1959), the differences between the correlations with CELEXwf and SUBTLEX were significant for all three RTs ($t(452) = -3.12$, $t(452) = -2.14$, $t(452) = -3.68$ respectively) and approached significance for the PEs ($t(452) = -1.65$, $t(452) = -1.70$, $t(452) = -2.21$ respectively). This presumably has to do with the small size of the Celex corpus and with the fact that it is the oldest corpus.
2. The Leipzig, the dlexDB, and the Google frequencies are very similar.
3. The Google frequencies are not much better than Leipzig or dlexDB despite the large differences in size. In line with the English findings, the Google frequencies limited to the books published between 2000 and 2009 are better than the Google frequencies based on the entire corpus.
4. The SUBTLEX-DE frequencies are always the best predictor, although the difference is not statistically significant for all comparisons. For instance, for the average RT across the three studies $r = .658$ for SUBTLEX-DE, against $r = .614$ for Google00 ($t(452) = 1.77$) and $r = .604$ for dlexDBwf ($t(452) = 2.16$).
5. Lemma frequencies tend to be less good predictors than word form frequencies.

These findings are in line with those of other languages. The superiority of the SUBTLEX frequencies remains when they are used in a polynomial regression analysis to capture the non-linearities of the frequency curve (Baayen et al., 2006; Balota et al., 2004; Keuleers et al., 2010b; Figure 1). For instance, when polynomials of the third degree are used, SUBTLEX-DE explains 37.1% of the variance in RT1, against 30.8% explained by dlexDBwf and 33.8% explained by Google00. The difference with CELEXwf (27.7%) is even larger, which is a concern when one takes into account that many variables in lexical decision time explain only a small percentage of unique variance.

Dataset 2: 451 words presented in a lexical decision task

The second dataset used to validate the various word frequency measures involved 451 words. These words were presented in a single standard lexical decision experiment with legal non-words (see below for a description of the material).

Participants. Participants were 40 undergraduate students from the Freie Universität Berlin. All had normal or corrected to normal vision and were naïve with respect to the research hypothesis. They took part on a voluntary basis.

Stimuli. The word stimuli consisted of 451 words, which were selected to examine differential effects of orthographic vs. phonological initial syllable frequency in German (see Conrad, Grainger, & Jacobs, 2007 for such differential effects in French). Nonwords were easily pronounceable letter strings matched to the words on length, syllable number and initial capitals.

Method. Before the test session 10 practice trials were administered. The test session itself consisted of 902 trials presented in lowercase (Courier 24) with initial capitals for nouns. The time line of a trial was: Items were presented in the center of the screen after a fixation cross (500ms) until responses were given, followed by an intertrial interval of 500 ms. Participants responded with their right hand to the word trials and with their left hand to the non-word trials. Each participant got a different, random permutation of the stimulus list.

Results. The mean reaction time to the words was 789 ms with an error rate 14.6%. Performance was less well than in the first dataset, despite the fact that the words were of similar frequency according to the Google measure (mean $\log_{10}(\text{Google per million} + 1/30840) = .30$, $SD = .809$).

Insert Table 3 about here

3 shows the results, which in all aspects converge with those of Table 2.

Insert Table 3 about here

Again, the correlations between RT or PE and SUBTLEX were significantly higher than the correlations with CELEXwf ($t(448) = 3.23$ and $t(448) = 3.19$, respectively). The differences

with the best contender, Leipzig, failed to reach significance (RT: $t(448) = 1.32$; PE: $t(448) = .07$).

Dataset 3: 2,152 compound words presented in two lexical decision tasks

Analysis of English data indicated that SUBTLEX frequencies did particularly well for short words and that written frequencies did better for longer words (Brysbaert & New, 2009, Table 5). Given that German has more long words than English (because compound words are written conjointly), it is important to also have information about long compound words. Therefore, we used a third dataset consisting entirely of this type of words (Böhl, 2007).

Participants. Participants were two groups of 16 undergraduate students from Westfälische Wilhelms-Universität Münster. All had normal or corrected to normal vision and were naïve with respect to the research hypothesis. They took part on a voluntary basis and received course-credit or were paid 10 € for participation.

Stimuli. The word stimuli consisted of 2,152 compound words. They had an average frequency of $\log_{10}(\text{Google per million} + 1/30840) = -.89$ (equivalent to .1 per million), $SD = .824$. The words were presented in two different lexical decision experiments with different non-words. In Experiment 1, pseudowords were constructed from the original compounds by changing the initial, medial or final phoneme in the first or second constituent. In Experiment 2, pseudowords were non-existing compounds consisting of two existing constituents, e.g., Sahnetisch (cream table). Compound words and pseudowords were equally distributed over two lists consisting of 2,152 stimuli, half of which were compound words and half pseudowords.

Method. Each participant received one list which was further divided into eight blocks consisting of about 269 trials. The eight blocks were distributed over two sessions. Before each block started, the participants received 15 warming-up trials. Each participant got a different, random permutation of the stimulus list.

Eye movements were tracked during the experiment. Before the experiment the eye-tracker was calibrated. At the start of each trial, a fixation point was presented at the left margin of the screen (centered 100 pixels to the right of the left screen margin). This fixation point had to be fixated before each trial started. After successful fixation the stimulus appeared in the centre of the screen, 100 pixels (2.2°) to the right of the fixation point. All stimuli were presented in Courier (font size 26). Participants responded with their dominant hand to the word trials and with their non-dominant hand to the non-word trials.

Results. Mean reaction time of Experiment 1 was $RT1 = 797$ ms ($SD = 154$) and $PE1 = 9.2\%$ ($SD = 15.9$); in Experiment 2 the values were $RT2 = 985$ ms ($SD = 223$) and $PE2 = 31.2\%$ ($SD = 21.3$). The use of pseudocompounds in Experiment 2 clearly made the task more difficult, because participants had a hard time deciding which compounds were found in the German language and which not. The correlation between the RTs was .45; that between the PEs was .52, which is considerably lower than in the previous datasets and is most likely due to the small number of observations per word (eight). The reduced reliability of the dependent variables means that the correlations with the frequency measures will be lower as well. Table 4 shows the results.

Insert Table 4 about here

Despite the concerns we had about the usefulness of SUBTLEX for longer words, it again turned out to be the best predictor, particularly for RTs (which tend to be the most important dependent variable in psycholinguistics). The correlation between RT1 and SUBTLEX was significantly higher than that with CELEXwf ($t(1953) = 6.77$), Google00 ($t(1953) = 2.83$), Leipzig ($t(1953) = 2.77$), and dlexDB ($t(1953) = 2.00$). These values were calculated on the words that were recognized by at least two thirds of the participants. The same was true for RT2 ($t(1299)$ -values respectively of 5.13, 2.66, 2.54, and 3.23). The difference in correlation with PE1 were smaller and only significant for CELEXwf ($t(2149) = 3.56$), but not for the other measures, which sometimes had higher correlations (e.g., Leipzig: $t(2149) = -1.20$). Of all frequency measures, SUBTLEX correlated most with PE2 and the difference was significant for CELEXwf ($t(2149) = 7.85$), dlexDBwf ($t(2149) = 5.86$), Google00 ($t(2149) = 3.12$), but not for Leipzig ($t(2149) = 1.42$).

Differences in language register between SUBTLEX-DE and dlexDB

In the previous analyses we replicated the now robust findings that word frequencies based on some 20 million words from popular films and television series outperform estimates based on much larger written corpora. We speculated that this has to do with the language register tapped by the measures: Social interactions in films vs. descriptions and explanations in written corpora (certainly in non-fiction works). One way to get a better idea of what this difference implies is to see for which words the frequency measures differ most.

An easy way to do this is to predict one frequency measure on the basis of the other, and look at the residual scores to decide which words are most overestimated or underestimated. Table 5 shows the outcome of such an analysis for the three stimulus sets we used, when SUBTLEX and dlexDBwf are compared to each other. More than any other analysis, this table illustrates that subtitle frequencies tap more into the informal, social language with words such as “kiss, sex, monster, say, rotten pig (as description of a man), and freckle”. In contrast, the written sources currently available deal more with descriptions and explanations (“dizziness, material, Köhler (name of former German president; also means charcoal burner), accumulation, state parliament, and first principle”).

Insert Table 5 about here

Conclusions

The analyses presented in this paper confirm the findings of Brysbaert and New (2009) and Keuleers et al. (2010b) for the German language: There are considerable quality differences in the various word frequency measures available and these can be investigated by correlating the frequency measures with word processing times, in particular lexical decision times.

On the basis of our analyses, it looks like the Celex frequencies, which were of such importance in the pre-digital period, have had their best time. Because of the small and dated corpus on which they are based, the percentage of variance explained in lexical

decision times is consistently lower than the percentage explained with more recent estimates. The Celex database, however, still is a rich source of information about, for instance, the morphology of words.

The three remaining frequency measures based on written sources, Leipzig, dlexDB, and Google Ngram Viewer, are of similar quality, at least when the Google measure is limited to the books published after 2000 (Tables 2-4). This illustrates that the differences in corpus size (49 million words, 100 million words, 6.1 billion words) do not matter from a certain size on. Unfortunately, the Leipzig measure is not easily available, as only one word at a time can be entered into the website (<http://wortschatz.uni-leipzig.de/>). Additionally, the website only gives an absolute count, making it difficult to derive frequencies per million (given that the corpus is constantly updated). On the other hand, the Leipzig website provides a lot of extra information about each word (e.g., about the meaning), which may be of great interest to psycholinguistics.

The dlexDB frequencies are more easily available (www.dlexdb.de). In addition, there is information about lemma frequencies and the frequencies of the various syntactic roles taken by the words. Finally, the website also contains information about the orthographic and phonological relationship of each word to other words (e.g., number of neighbors, Levenshtein Distance to other words, etc.).

The Google Ngram frequencies can easily be downloaded from the website (<http://ngrams.googlelabs.com/datasets>; go under German 1-grams). Their main advantage is the size of the corpus, which means that more detailed frequency information can be

obtained about very low-frequency words. Also, the separation as a function of the year in which the books were published may provide researchers with interesting information as to the shelf life of word frequency estimates.

As in other languages, frequencies based on film and television subtitles are the best predictor of word processing efficiency in psychological experiments. Several factors are likely to be involved in this superiority. First, participants in psychological studies may have watched more television in their life than spent time reading. Second, the language on television may be closer to everyday language use than the language used in written texts, certainly if the latter contain a lot of non-fiction works. Third, the sample of subtitle files we were able to download from the internet mainly contained popular films and television series (i.e., the ones participants are likely to have watched). This is different from a corpus that contains everything published from a particular source (e.g., a newspaper or a magazine). Such a corpus is more likely to include works read by only a few specialists. Finally, because children are more likely to watch television than to read books, subtitles may better capture language use in childhood than written sources.

It may be interesting to notice that nothing prevents researchers from building a written corpus based on the same principles. Because of the large costs associated with the compilation of word frequency lists in the past psychologists had to do with whatever was available. However, at present this situation is changing rapidly, certainly when the corpus can be as small as 20-50 million words. Due to the massive availability of digital sources, it now becomes feasible to assemble a corpus that is representative for the type of words participants in psychology experiments have been exposed to (e.g., popular books, much

read newspapers and magazines, often visited internet sites and discussion groups, much used school books, and so on). Our analyses suggest that such a written corpus could be a very useful addition to a subtitle corpus.

To some extent it is surprising that subtitle-based frequencies are doing so well to predict lexical decision times to printed words for languages such as German and French, given that films and television programs are rarely subtitled in these languages (they usually get a voice-over in the new language). This is different from languages such as Dutch and Greek where a large part of television programs are subtitled and people are used to reading these subtitles. The fact that SUBTLEX frequencies are doing so well in English, French, and German suggests that auditory word exposure contributes to the efficiency of visual word processing, in line with interactive models of word recognition (e.g., Ziegler, Petrova, & Ferrand, 2008). This auditory exposure arguably not only comes from television, but also from everyday social interactions, and maybe from self-thought as well. Because of the widespread use of subtitles in the Greek- and Dutch-speaking regions, Dimitropoulou et al. (2010) hypothesized that subtitle frequencies may have an extra advantage for these languages, because they also make up a large part of the texts read by undergraduate students.

Finally, it might be argued³ that the quality differences between the frequency measures, discussed here, have rather limited effect on research in which a sample of low-frequency words (e.g., of 1-2 pm) is compared to a sample of high frequency words (of 50-10,000 pm). This is true, as long as the distinction between the two samples is a crude one. As Figure 1

³ As was done by one of our reviewers.

shows, the weak part of the research will be the low-frequency words. Misestimates of the frequencies of these words can easily lead to differences of 50-100 ms, resulting in a large or a small frequency effect. The estimates of low-frequency words are likely to be an even bigger issue when researchers want to match their stimuli on word frequency. Then, small differences between the “matched” lists in the low-frequency range can easily lead to spurious results (particularly when the frequencies were not matched on log frequency; see again Figure 1). Indeed, a look at recent publications indicates that authors use word frequency estimates more often to match their stimuli than to investigate the frequency effect itself (which is already well-established). A search through the latest issues of *Experimental Psychology* confirms this impression. Recent examples of articles making use of word frequency estimates are Coane and Balota (2011) who matched four types of primes on frequency, Dunabeitia, Perea, and Carreiras (2010) who matched cognate and noncognate words on frequency, Hartsuiker and Notebaert (2010) who matched pictures with low and high name agreement on word frequency, Stein, Zwickel, Klitzmantel, Ritter, and Schneider (2010) who matched neutral, negative, and taboo words on frequency, Duyck and Warlop (2009) who matched words from first and second language on frequency, and so on. All these authors assumed that the frequency estimates they were using were valid ones (and the best available).

Availability of the SUBTLEX-DE word frequencies

Because word frequency measures are of interest only if they can be accessed easily by researchers, we made the SUBTLEX-DE word frequencies available in easy-to-use spreadsheet files that can be downloaded from the website <http://crr.ugent.be/subtlex-de/>. The first type of file (there are various formats) is the raw output we obtained from the word

counting, taking into account the differences between uppercase and lowercase letters. It contains 377,524 lines of words with their frequency counts (FREQcount, on a total of 25,399,098 tokens). For each entry, there is also information about whether the word was accepted by the Igerman98 spell checker (1) or not (0). This spell checker is used in many German text processing packages (see <http://freshmeat.net/projects/igerman98>). It provides interesting information to filter out typos and foreign names.⁴ However, the spell checker tends to miss low-frequency morphological complex words, which may be of interest to researchers. It also tends to accept special characters that could be punctuation marks (e.g. '/'). The raw file is an ideal file for checking various questions related to word frequencies.

The second file is a version with some basic cleaning. First, the frequencies are given independently of capitalization. In addition, the words are written with a capital if this spelling is more frequent in the subtitle corpus and with a lowercase if that is more frequent. Second, we excluded the entries that started with non alphabetical characters. Finally, we added the Google 2000-2009 frequencies to the file and excluded all entries that were not observed in the Google corpus. This resulted in 190,500 remaining entries. The addition of the Google frequencies gives researchers extra information about the stimuli. This is particularly useful for words with a very low frequency (given that the Google frequencies are based on a corpus of 6.05 billion words).

Insert Figure 2 about here

⁴ The authors thank Emmanuel Keuleers for suggesting this possibility.

Figure 2 gives a screenshot of the first entries in the cleaned SUBTLEX-DE version (ranked from most frequent to least frequent). The information given in the different columns is as follows:

- **Word** : This is the word for which the frequencies are given. If the word in the subtitle corpus most of the time starts with a capital, it is written with a capital (Ich, Sie, Was, ...). If it mostly starts with a lowercase, it is written that way (das, ist, du, ...).
- **WFfreqcount** : This is the number of times the word as written in column 1 is encountered in the subtitle corpus.
- **Spell check** : The numbers in this column indicate whether the word was accepted by the Igerman98 spell checker (1) or not (0). This is particularly interesting to avoid uninteresting entries when one wants to generate word lists fulfilling particular constraints.
- **CUMfreqcount**: This is the number of times the word is encountered in the subtitle corpus, independently of letter case. It is the value on which the SUBTLEX frequency and lgSUBTLEX are based.
- **SUBTLEX**: This is the frequency per million based on CUMfreqcount (i.e., it equals $\text{CUMfreqcount} / 25.399$). This is the value to report in manuscripts, because it is a standardized value independent of the corpus size. The value is given up to two decimal places in order not to lose information (notice the use of the comma as the decimal sign, which is the standard in German speaking countries).
- **LgSUBTLEX**: This is $\log_{10}(\text{CUMfreqcount}+1)$. It is the value to use when you want to match stimuli in various conditions. When a stimulus is not present in the corpus,

lgSUBTLEX gets a value of 0. If you want to express the value as $\log_{10}(\text{frequency per million})$, simply subtract $\log_{10}(25.399) = 1.405$ from lgSUBTLEX.

- **Google00**: This is the number of times the word as written in column 1 is encountered in the Google 2000-2009 Books corpus.
- **Google00cum**: This is the number of times the word appears in the Google 2000-2009 Books corpus, independently of the case of the first letter.
- **Google00pm**: This is the Google frequency per million words. It is obtained by applying the equation $\text{Google00cum} / 6050.356524$.
- **lgGoogle00**: This value equals $\log_{10}(\text{Google00cum}+1)$. If you want to express the value as $\log_{10}(\text{frequency per million})$, simply subtract $\log_{10}(6050.4) = 3.872$ from lgGoogle00.

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Table 1: Contribution of the different word variables in the Elexicon Project towards explaining the variance in lexical decision times. Outcome of a stepwise multiple regression analysis. Stimuli cleaned for genitive forms, words that had no values of Orthographic Levenshtein Distance (OLD) or Number of morphemes, and stimuli that had no frequency in any database ($N = 38,436$)

<i>Variable</i>	<i>R^2</i>
Word frequency (log SUBTLEX)	.405
Orthographic Levenshtein Distance (OLD)	.534
Number of syllables	.546
All	.566

Table 2: Correlations between the dependent variables from the first dataset (3 experiments) and the different frequency measures. All frequencies are log transformed. Correlations are based on 455 words and are all significant.

	RT1	RT2	RT3	PE1	PE2	PE3
CELEXwf	-.506	-.493	-.524	-.286	-.317	-.294
CELEXlem	-.478	-.453	-.480	-.280	-.300	-.260
Leipzig	-.534	-.507	-.579	-.306	-.369	-.371
dlexDBwf	-.550	-.522	-.581	-.352	-.362	-.318
dlexDBlem	-.521	-.496	-.521	-.350	-.356	-.292
Google	-.542	-.500	-.546	-.325	-.364	-.323
Google80	-.527	-.483	-.542	-.300	-.342	-.322
Google90	-.544	-.497	-.559	-.305	-.351	-.336
Google00	-.575	-.523	-.586	-.328	-.381	-.369
SUBTLEX	-.602	-.561	-.634	-.346	-.378	-.374

Table 3: Correlations between the dependent variables from the second dataset and the different frequency measures. All frequencies were log transformed. Correlations are based on 451 words and are all significant.

	RT	PE
CELEXwf	-.513	-.456
CELEXlem	-.536	-.443
Leipzig	-.576	-.557
dlxDBwf	-.566	-.530
dlxDBlem	-.519	-.493
Google	-.507	-.486
Google80	-.482	-.467
Google90	-.497	-.476
Google00	-.538	-.524
SUBTLEX	-.612	-.559

Table 4: Correlations between the dependent variables from the third dataset and the different frequency measures. All frequencies were log transformed. Correlations are based on all trials for the PEs, on N = 1,956 for RT1 (PE1 < 33%) and N= 1,302 for RT2 (PE2 < 33%); they are all significant ($p < .01$).

	RT1	RT2	PE1	PE2
CELEXwf	-.202	-.244	-.216	-.306
CELEXlem	-.213	-.268	-.232	-.330
Leipzig	-.293	-.317	-.309	-.431
dllexDBwf	-.303	-.294	-.268	-.346
dllexDBlem	-.312	-.289	-.282	-.358
Google	-.234	-.245	-.232	-.287
Google80	-.224	-.240	-.241	-.301
Google90	-.243	-.268	-.254	-.331
Google00	-.288	-.311	-.303	-.399
SUBTLEX	-.344	-.375	-.288	-.454

Table 5: The thirty words from the three datasets for which the SUBTLEX and DLEXwf frequencies differ most.

	Dataset1	Dataset2	Dataset3
SUBTLEX > dlexDBwf	Kuss	Monster	Mistkerl
	Sex	sage	Sommersprosse
	Vampir	lese	Notstrom
	lecker	Koma	Wunderkerze
	töten	Virus	Autokino
	anlügen	Viper	Rauchbombe
	ableben	Wade	Ohrfeige
	Halt	Party	Waffeleisen
	abhauen	Ahnung	Schuljunge
	Agent	Foto	Mistelzweig
SUBTLEX < dlexDBwf	Taumel	Köhler	Landtag
	materiell	Häufung	Grundlage
	Affekt	Prägung	Generalleutnant
	Weltmeer	Tegel	Landrat
	Unrat	Bayer	Hofrat
	Sanftmut	Senkung	Parteitag
	bedächtig	Linde	Justizrat
	abgründig	Entgelt	Stahlhelm
	Exotik	Koloß	Stadttheater
	verwaist	Reede	Luftwaffe

Figure 1: The word frequency RT-curve for the word stimuli in the Dutch Lexicon Project, the English Lexicon Project, and the French Lexicon Project. Stimulus frequencies were obtained from SUBTLEX-NL, SUBTLEX-US, and Lexique 3.55 (New et al., 2007) respectively. They varied from .02 to nearly 40,000 per million words (pm). Circles indicate the mean RT per bin of .15 log10 word-frequency pm; error bars indicate 2*SE (bins without error bars contained only one word). The large error bars on the right side are due to the fact that there are very few high-frequency words in these bins. Source: Keuleers et al., 2010b, Figure 4 (available at http://94.236.98.240/language_sciences/10.3389/fpsyg.2010.00174/abstract; open access).

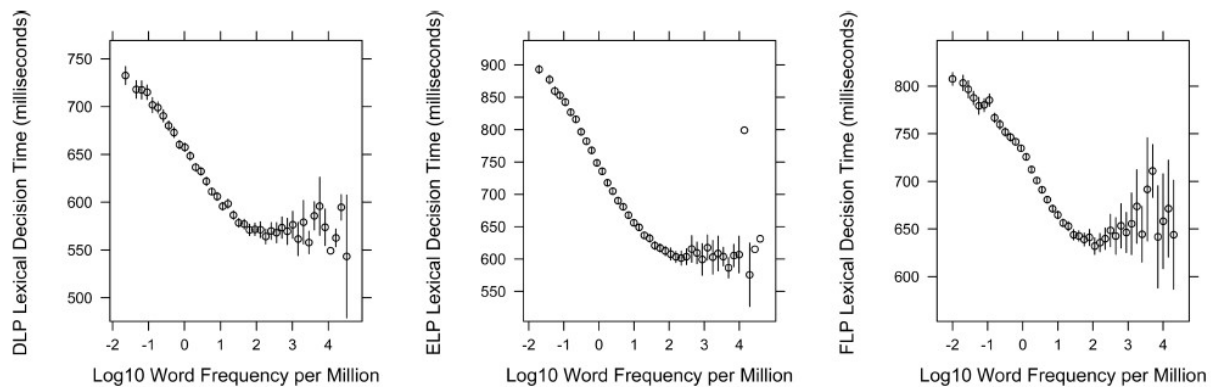


Figure 2: Screenshot of the first entries in SUBTLEX-DE cleaned version

A1		f _x Word								
	A	B	C	D	E	F	G	H	I	J
1	Word	Wffreqcount	spell-check OK	CUMfreqcount	SUBTLEX	lgSUBTLEX	Google00	Google00cum	Google00pm	lgGoogle00
2	Ich	489637	1	930449	36633,29	5,969	4506068	15190524	2510,68246	7,182
3	Sie	366870	1	566657	22310,21	5,753	13823808	35381018	5847,75754	7,549
4	das	326943	1	515937	20313,28	5,713	45066596	56780837	9384,70928	7,754
5	ist	475043	1	507382	19976,46	5,705	43154392	43918032	7258,75109	7,643
6	du	342560	1	506555	19943,90	5,705	1923152	2660677	439,75541	6,425
7	nicht	422376	1	443722	17470,06	5,647	38334374	39503887	6529,18334	7,597
8	die	326305	1	409915	16139,02	5,613	170961824	204236386	33756,09110	8,310
9	es	261560	1	377458	14861,14	5,577	23614920	30887546	5105,07866	7,490
10	und	255151	1	367531	14470,29	5,565	157099890	160198542	26477,53754	8,205
11	der	235950	1	285513	11241,11	5,456	200767752	213886850	35351,11512	8,330
12	Was	146767	1	280565	11046,30	5,448	1944049	8387327	1386,25335	6,924
13	wir	166009	1	280268	11034,61	5,448	7409855	9602120	1587,03375	6,982
14	zu	229130	1	233923	9209,93	5,369	59472403	61004779	10082,84037	7,785
15	ein	185794	1	218031	8584,24	5,339	28480540	32735558	5410,51719	7,515
16	er	121771	1	214680	8452,30	5,332	15705428	19619934	3242,77320	7,293
17	in	182149	1	204750	8061,34	5,311	101411207	113074431	18688,88727	8,053
18	mir	158541	1	166463	6553,92	5,221	3164242	3303307	545,96898	6,519
19	mit	152783	1	163704	6445,29	5,214	39910825	42744231	7064,74583	7,631
20	Ja	105501	1	151904	5980,71	5,182	469720	2529820	418,12743	6,403